

Clinical Risk Assessment Report: Patient 77

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 35.9% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.62x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +5.4%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 8.89 [Top <1%] (+5.4%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.45 [Avg (71th %ile)] (+2.5%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 3.04 [Top <1%] (+2.4%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.6 [Avg (63th %ile)] (+1.9%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.28 [Avg (31th %ile)] (+1.7%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: 4.36 [Top <1%] (+1.6%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.18 [Avg (31th %ile)] (+1.4%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: 7.19 [Top <1%] (+0.7%)**
Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

